

Q. Given $3 \leq \sqrt{2x+5} < 6$

#1

We first break the inequality into two part:

$$3 \leq \sqrt{2x+5} \quad \text{and} \quad \sqrt{2x+5} < 6$$

The radicant of both inequality must be non-negative

$$\therefore 2x+5 \geq 0 \Rightarrow x \geq -\frac{5}{2}$$

i) for $3 \leq \sqrt{2x+5}$, the critical points will be:

$$3 = \sqrt{2x+5} \Rightarrow (3)^2 = (\sqrt{2x+5})^2 \Rightarrow x = \underline{2}$$

$\therefore$ Critical points are $x=2$ and $x=-\frac{5}{2}$.

Testing the points:

$x < -\frac{5}{2}$	$-\frac{5}{2} < x < 2$	$x > 2$
Take $x = -3$	Take $x = 0$	Take $x = 3$
$3 \leq \sqrt{-1}$	$3 \leq \sqrt{5}$	$3 \leq \sqrt{11}$
undefined	Not true	True.

N.B

AG $x=2$
 $3 \leq \sqrt{9} \Rightarrow 3 \leq 3$ so true, then
 (2) is also part of solution then

$\therefore$ the solution set of $3 \leq \sqrt{2x+5}$
 is $\underline{[2, \infty)}$

ii) for $\sqrt{2x+5} < 6$, the critical points

$$(\sqrt{2x+5})^2 = (6)^2 \Rightarrow 2x+5 = 36 \Rightarrow x = \frac{31}{2}$$

$$\therefore x = -\frac{5}{2} \quad \text{and} \quad x = \frac{31}{2}$$

Testing the points

$x < -\frac{5}{2}$	$-\frac{5}{2} < x < \frac{31}{2}$	$x > \frac{31}{2}$
$x = -3$	Take $x = 0$	Take $x = 17$ or 16
$3 \leq \sqrt{2x+5}$	$\sqrt{5} < 6$	$\sqrt{2(17)+5} < 6$
$3 \leq \sqrt{-1}$	True	$\sqrt{39} < 6$
undefined		Not True.

$\therefore$ The Combined Solution
 for $3 \leq \sqrt{2x+5} < 6$ is

$$\underline{\underline{2 \leq x < \frac{31}{2} \quad \text{OR} \quad [2, \frac{31}{2})}}$$

The solution set for this is $[-\frac{5}{2}, \frac{31}{2})$

Given a function q

2

(b) Given $10 - \sqrt{2x+7} \leq 3$

The radicand must be nonnegative, meaning it must not be negative.

$$\text{ie } 2x+7 \geq 0$$

$$x \geq -\frac{7}{2}$$

After that, we need to find the critical points.

$$\Rightarrow 10 - \sqrt{2x+7} = 3$$

$$(\sqrt{2x+7})^2 = (7)^2 \Rightarrow 2x+7 = 49 \Rightarrow x = \frac{21}{2}$$

$\therefore$ The critical points are $x = -\frac{7}{2}$ and $x = 21$.

Testing the critical points:

$x < -\frac{7}{2}$	$-\frac{7}{2} < x < 21$	$x > 21$
Take $x = (-4)$	Take $x = 0$	Take $x = 22$
$10 - \sqrt{2x+7} \leq 3$	$10 - \sqrt{2x+7} \leq 3$	$10 - \sqrt{2x+7} \leq 3$
$10 - \sqrt{-5} \leq 3$	$10 - \sqrt{7} \leq 3$	$10 - \sqrt{51} \leq 3$
Undefined.	Not True.	True ✓
X	X	

At $x = 21$

$$\Rightarrow 10 - \sqrt{2(21)+7} \leq 3$$

$$10 - \sqrt{49} \leq 3$$

$$10 - 7 \leq 3$$

$3 \leq 3$ True, Then 21 is also part of solution.

The solution set of $10 - \sqrt{2x+7} \leq 3$

$$\text{is } \{x \mid x \in \mathbb{R}, x \geq 21\} \text{ OR } [21, \infty)$$

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© Given $\sqrt{2x+9} - \sqrt{9+x} > 0$

#3

- The radicand of both radicals must be non-negative.

$\therefore 2x+9 \geq 0$ and $9+x \geq 0$

$\therefore x \geq -9/2$ and $x \geq -9$

Therefore the combined domain is $x \geq -9/2$ of the radicand

- Recall that the critical points are found with an inequality symbol replaced with an equal sign.

$\therefore \sqrt{2x+9} = \sqrt{9+x}$

$(\sqrt{2x+9})^2 = (\sqrt{9+x})^2$

$2x+9 = 9+x$

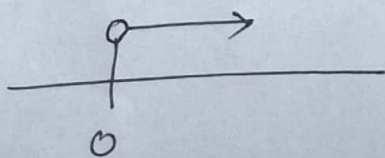
$x = 0$

- Therefore critical points are $x=0$ and $x = -9/2$

- Now let's test our critical points;

$x < -9/2$	$-9/2 < x < 0$	$x > 0$
let $x = -5$	let $x = -1$	let $x = 1$
$\sqrt{2(-5)+9} - \sqrt{9-5} > 0$	$\sqrt{7} - \sqrt{8} > 0$	$\sqrt{11} - \sqrt{9} > 0$
UNDERMINED!	NOT TRUE	TRUE ✓

Therefore Solution Set is $x > 0$ NB: 0 NOT INCLUDED PROVE!



Solution Set $(0, \infty)$

Ans.

[Signature]

④ Given $\sqrt{2} - \sqrt{x+6} \leq -\sqrt{x}$

#4

— The radicands must be non-negative.

$\therefore x \geq 0$ (for $\sqrt{x}$)

$x+6 \geq 0 \Rightarrow x \geq -6$

Therefore the combined domain of the radicands is $x \geq 0$

— Now let's find the critical points.

$\Rightarrow \sqrt{2} - \sqrt{x+6} = -\sqrt{x}$

$\Rightarrow \sqrt{2} = \sqrt{x+6} - \sqrt{x}$

$\Rightarrow (\sqrt{2})^2 = (\sqrt{x+6} - \sqrt{x})^2$

$\Rightarrow 2 = x+6 - \sqrt{x(x+6)} - \sqrt{x(x+6)} + x$

$\Rightarrow 2 = 2x+6 - 2\sqrt{x(x+6)}$

$\Rightarrow x+2 = \sqrt{x(x+6)}$

Squaring once more we have.

$\Rightarrow (x+2)^2 = (\sqrt{x(x+6)})^2$

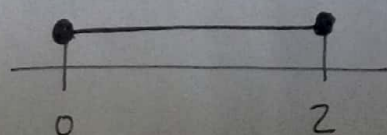
$\Rightarrow x^2 + 4x + 4 = x^2 + 6x \sim \text{ADDITIVE INVERSE}$

$\Rightarrow 2x = 4$

$\therefore \Rightarrow \underline{x = 4/2 = 2}$ is our critical point and $\underline{x = 0}$ is also our critical point. Now let's test.

$x < 0$	$0 < x < 2$	$x > 2$
let $x = -1$	let $x = 1$	let $x = 3$
$\sqrt{2} - \sqrt{-1+6} \leq -\sqrt{-1}$	$\sqrt{2} - \sqrt{7} \leq -\sqrt{1}$	$\sqrt{2} - \sqrt{9} \leq -\sqrt{3}$
UNDEFINED!	TRUE	NOT TRUE

Therefore the Satisfying Solution Set is $0 \leq x \leq 2 \Leftrightarrow [0, 2]$



with both 0 and 2 part of the solution.

Signature

e) Given $\sqrt{x-3} > \sqrt{x+4} - 1$

5

Recall; the radicands must be non-negative.

$$\therefore x-3 \geq 0 \Rightarrow x \geq 3$$

$$\text{Also } x+4 \geq 0 \Rightarrow x \geq -4$$

Therefore, the Combined domain of the radicands is $x \geq 3$

Now we find the Critical values.

$$\Rightarrow \sqrt{x-3} = \sqrt{x+4} - 1$$

$$\Rightarrow (\sqrt{x-3})^2 = (\sqrt{x+4} - 1)^2$$

$$\Rightarrow x-3 = x+4 - 2\sqrt{x+4} + 1$$

~ ADDITIVE INVERSE!

$$\Rightarrow 8 = 2\sqrt{x+4} \Rightarrow 4 = \sqrt{x+4}$$

Squaring again we have,

$$4^2 = (\sqrt{x+4})^2$$

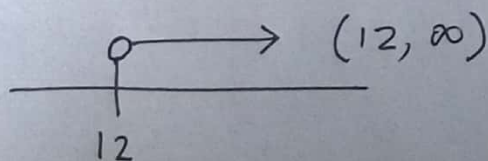
$$16 = x+4$$

$$\therefore x = 12 \text{ is our Critical point and } x = 3$$

Now let's test our Critical values.

$x < 3$	$3 < x < 12$	$x > 12$
let $x = 0$	let $x = 4$	let $x = 13$
$\sqrt{3} > \sqrt{4} - 1$	$\sqrt{1} > \sqrt{8} - 1$	$\sqrt{10} > \sqrt{17} - 1$
UNDEFINED!	NOT TRUE!	TRUE ✓

Therefore the solution set is $x > 12$ with 12 not part of the solution set.



Solution is $(12, \infty)$

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(f) Given $3 + \sqrt{2x-7} \leq 6$

#6

- The radicand must always be non-negative.

$\therefore 2x-7 \geq 0 \Rightarrow x \geq \underline{\underline{7/2}}$

- The Critical values are found as;

$$3 + \sqrt{2x-7} = 6$$

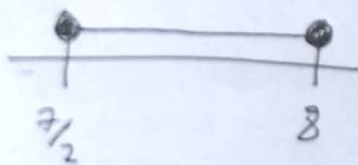
$$\Rightarrow \sqrt{2x-7} = 3 \Rightarrow (\sqrt{2x-7})^2 = 9 \Rightarrow 2x-7 = 9$$

$$\Rightarrow \underline{\underline{x = 8}}$$

Therefore the Critical values are $x = 8$ and $x = 7/2$. Now let's test

$x < 7/2$	$7/2 < x < 8$	$x > 8$
let $x = 0$	let $x = 4$	$x = 9$
$3 + \sqrt{-7} \leq 6$	$3 + \sqrt{1} \leq 6$	$3 + \sqrt{11} \leq 6$
UNDEFINED!	$\rightarrow$ TRUE $\checkmark$	NOT TRUE!

Therefore the Solution set is $\underline{\underline{7/2 \leq x \leq 8}}$



Solution Set $\underline{\underline{[7/2, 8]}}$

QUICK CHECK YOURSELF! Solve each of the following.

① $\sqrt{2x+5} < \sqrt{9+x}$

② $\sqrt{x+3} + \sqrt{x+7} > 4$

Ramin Kari